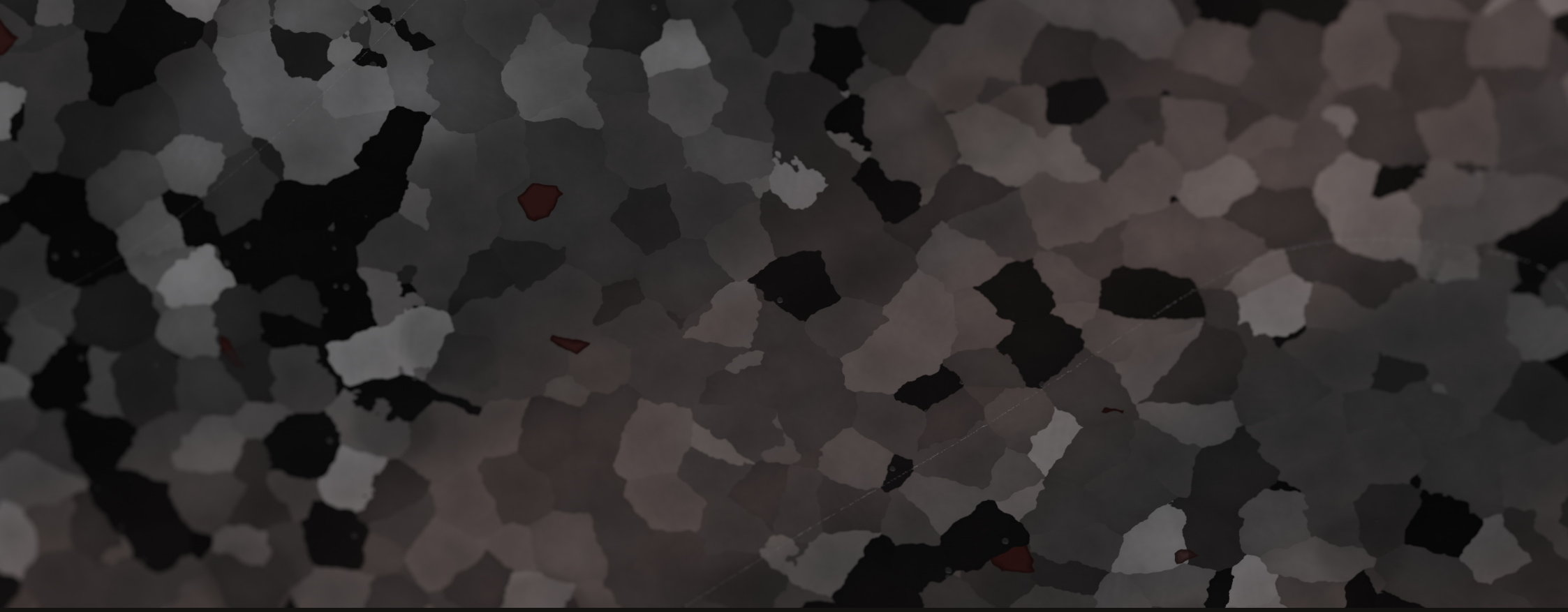


MATERIAL REPORT · V4 MACRO MICRODETAIL

Paradiso & Bahama Blue up close

Substance-Designer-style surface detail layered on the calibrated granite materials: crystal facets, mineral-specific polish, inclusions, polishing marks and finish-specific relief, checked under soft and grazing light.

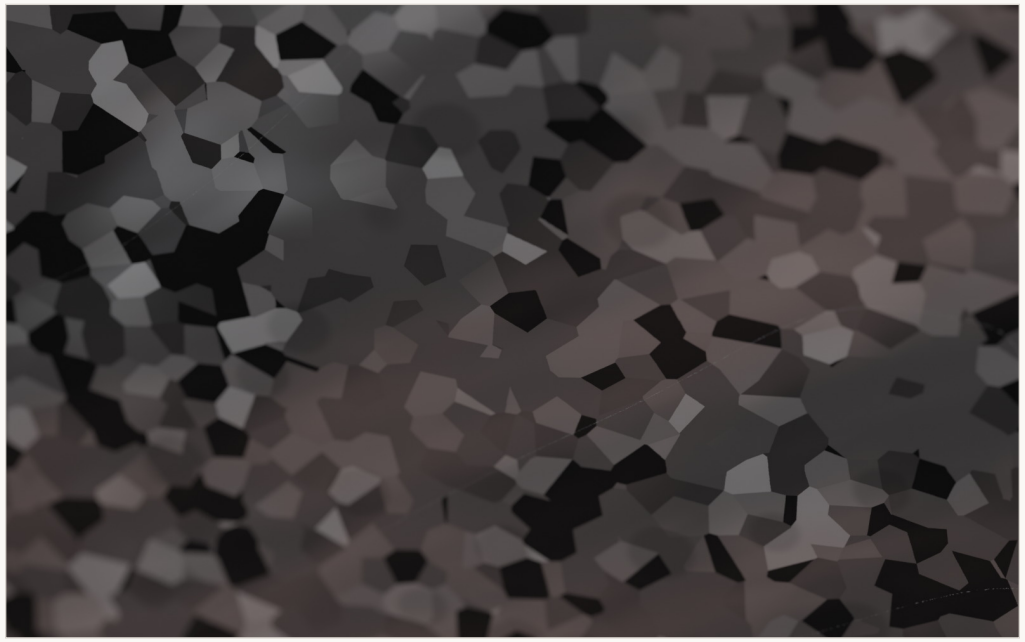
7	2	10–12	0
materials refined	separate micro graphs	controls per material	colour drift

Blender 5.2 LTS · Cycles · October 2026

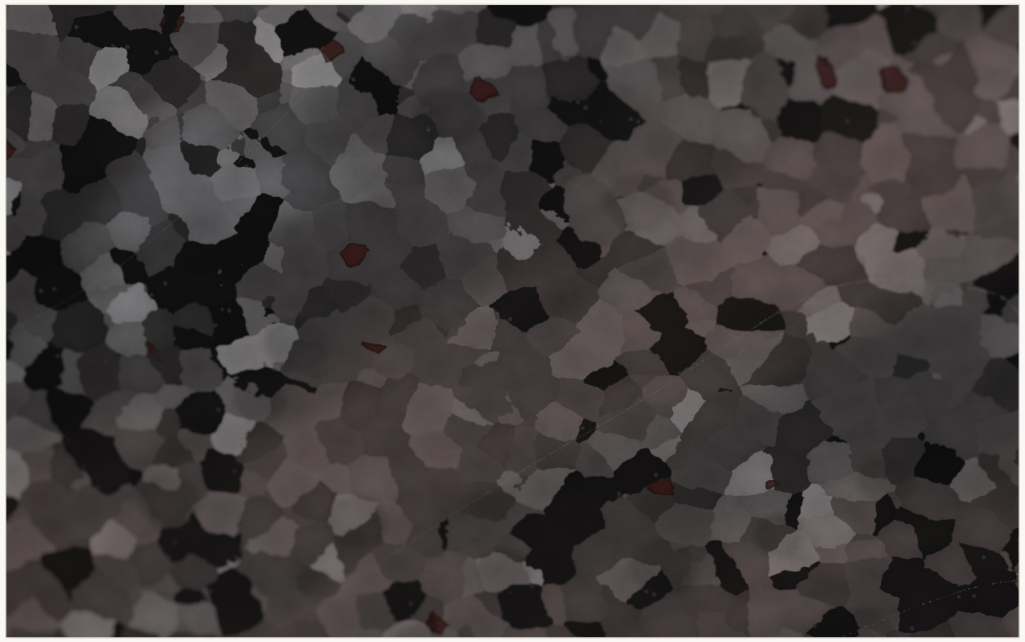
OVERVIEW

What changed and why

Paradiso 01, 40 mm field of view, soft light



BEFORE (V3) flat single-colour crystals, straight edges, one size



AFTER (V4) lobate crystals, zoning, garnets, mineral-specific polish

Problems at macro scale

- Crystals were flat Voronoi polygons: one colour, straight edges, all one size, so the stone read as camouflage.
- The polished-face bump made every crystal a small dome, which looks pitted under grazing light.
- Roughness was nearly uniform, so there was no mineral-to-mineral change in polish.

Approach (SD-style)

- Keep the calibrated pattern graph for colour and mineral layout.
- Add a separate micro graph per stone. It reads crystal id, edge and centre distance plus a dark-mineral mask, and outputs colour, roughness and a normal.
- Heights are in real metres (µm-scale), so Bump Strength 1 is physical.

Preserved

- Colours and mineral patterns: pattern albedo matches v3 (25 cm patch luminance, Paradiso 01: 0.275 vs 0.275, σ 0.122 vs 0.122; Bahama Blue 0.278 vs 0.278).
- Seeds, presets and the calibrated look at normal viewing distances.
- Originals kept as *_ORIG materials in the .blend; v3 scripts backed up.

METHOD

Material graph and physical scales

every material: Pattern → Microdetail → Shading

1 Pattern (calibrated)
GRANITE_<id> group
colour, mineral layers, seed
+ Grain Cell / Edge / Dist

→

2 Microdetail
MICRO_<id> group
polish, inclusions, relief,
facet normals (per stone)

→

3 Shading
Principled BSDF, IOR 1.54
+ displacement for
rock-pitched only

Paradiso micro graph

LAYER	SCALE / AMOUNT	EFFECT
Crystal facets	per crystal, 0.2° tilt	reflection flicker crystal to crystal
Crystal polish	roughness ±0.018	quartz / feldspar 0.05–0.09
Zoning + mottle	0.4 & 0.12 cell, ±14 %	volume inside crystals
Twin lamellae	0.13 mm, ~35 % feldspar	faint, irregular banding
Biotite	−1.5 µm, rough. +0.18	dull, slightly sunken mica
Plucked pits	0.18–0.4 mm, −10 µm	rare, only in biotite
Grain boundaries	15 µm, −0.7 µm	hairline micro-cracks
Garnets	0.5–1 mm, +0.8 µm	angular wine-red, dark rim, glossy
Polishing	scratches, ±6 µm / 7 cm	scratch sets, waviness, haze

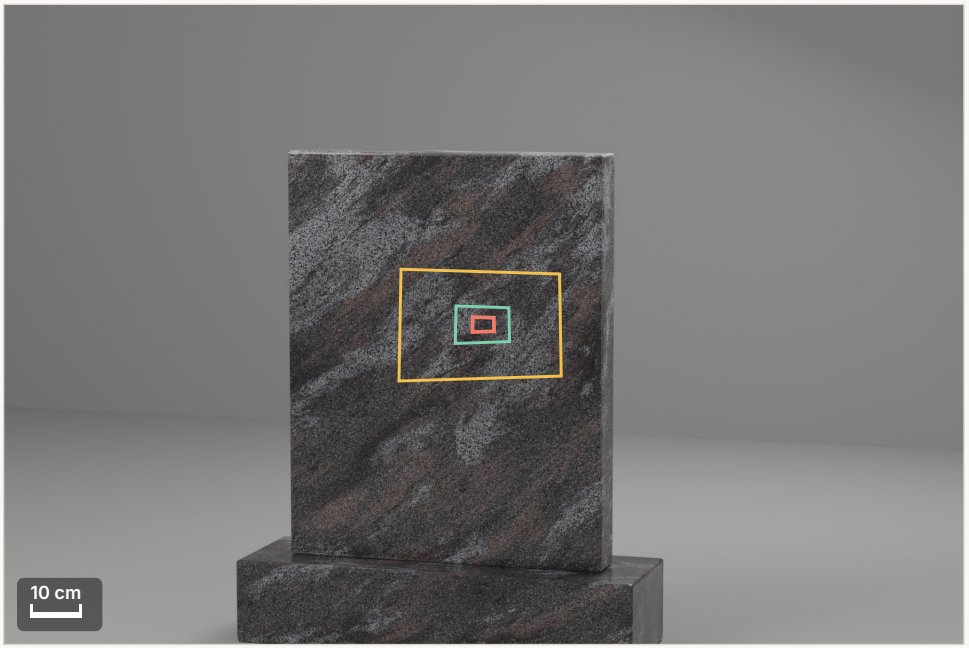
Bahama Blue micro graph

LAYER	SCALE / AMOUNT	EFFECT
Fine sparkle	0.15 mm, rough. ±0.012	replaces big facets
Facets	0.13° tilt	subtle (fine grain)
Fuzz + mosaic	0.5 cell, ±12 %	soft crystals, darker seams
Dark network	−0.8 µm, rough. +0.10	well bonded, nearly flush
Boundaries	8 µm, −0.5 µm	tight fabric
Plucked pits	rare (0.1)	almost none
Flamed spalls	1.8 mm, 0.3 mm steps	conchoidal craters, rough 0.8–0.9
Mica glints	roughness 0.22	sparkle on flamed edges
Polishing	same model, lighter	scratches, waviness, haze

SCALE CHECK

Paradiso: long shot to macro

same material, same seed, one camera axis · 100 mm lens



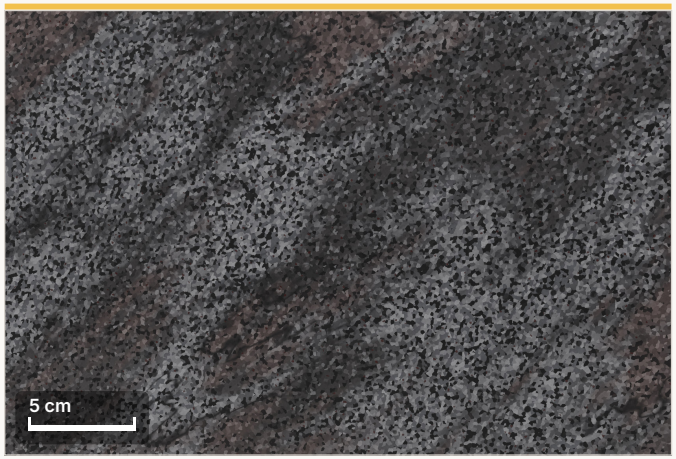
LONG SHOT headstone 62 × 78 cm on a 16 cm base, ~1.8 m wide frame

How to read this page

- All four images are separate renders of the same polished material and seed, taken along one camera axis. Only the distance changes (5.0 m, 0.83 m, 0.28 m, 0.11 m), so the pattern you see in the long shot is the same stone you see up close.
- The coloured outlines on the long shot mark exactly what each closer shot covers.
- Scale bars are measured from the render geometry at the centre of each frame.

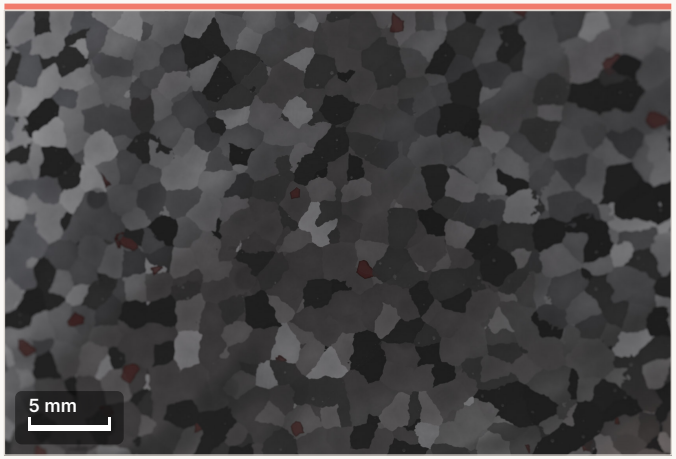
What to check

30 cm field	layering and mineral clusters
10 cm field	individual crystals become visible
4 cm field	crystal shapes and microdetail; mean crystal ≈ 1.5 mm



30 CM FIELD 31.2 cm across at the centre

10 CM FIELD 10.3 cm across at the centre

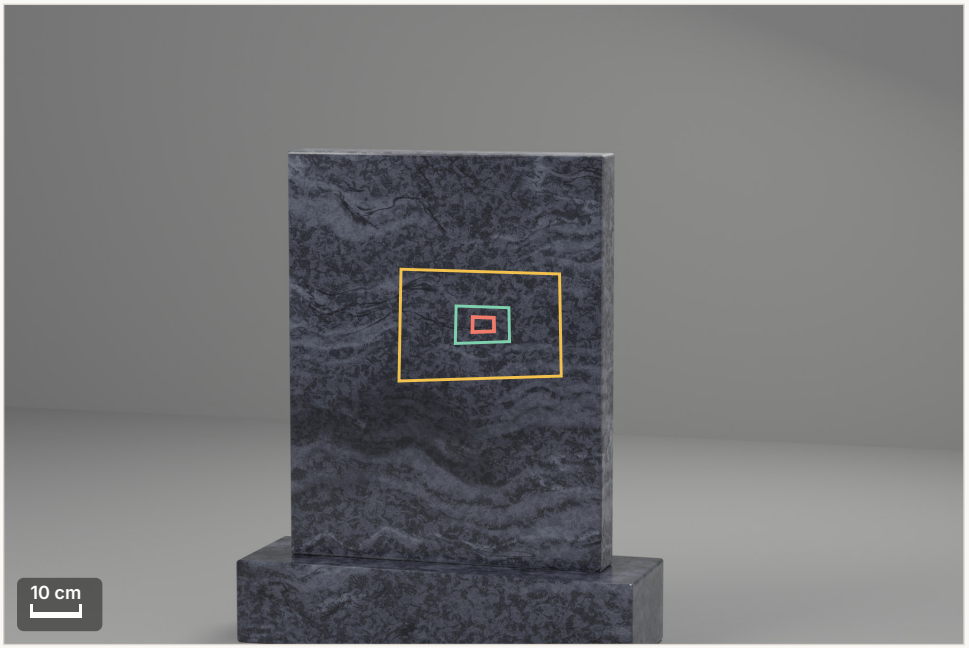


4 CM FIELD (MACRO) 4.1 cm across at the centre

SCALE CHECK

Bahama Blue: long shot to macro

same material, same seed, one camera axis · 100 mm lens



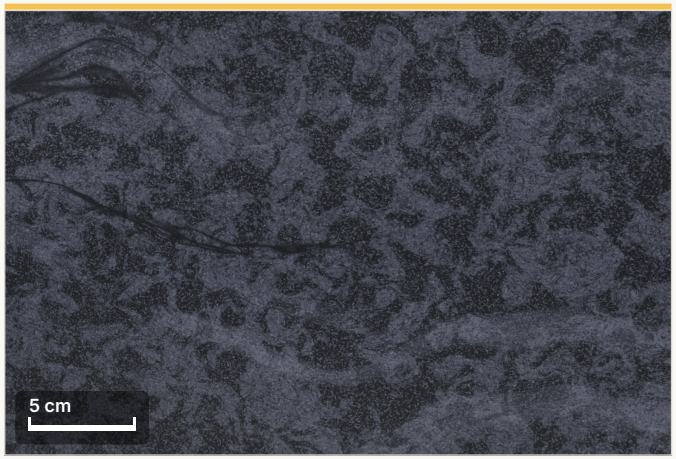
LONG SHOT headstone 62 × 78 cm on a 16 cm base, ~1.8 m wide frame

How to read this page

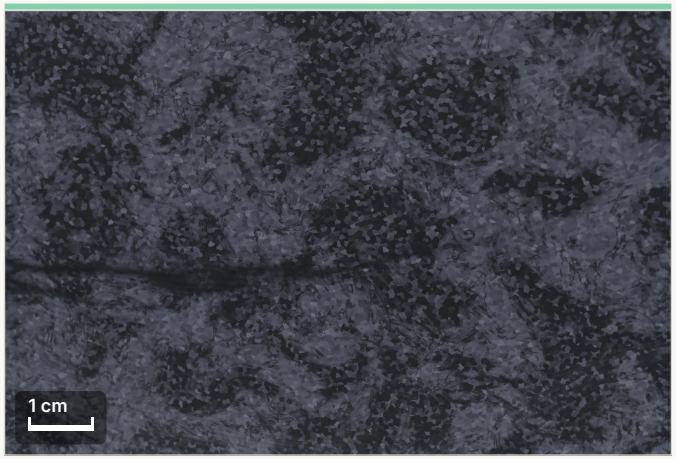
- All four images are separate renders of the same polished material and seed, taken along one camera axis. Only the distance changes (5.0 m, 0.83 m, 0.28 m, 0.11 m), so the pattern you see in the long shot is the same stone you see up close.
- The coloured outlines on the long shot mark exactly what each closer shot covers.
- Scale bars are measured from the render geometry at the centre of each frame.

What to check

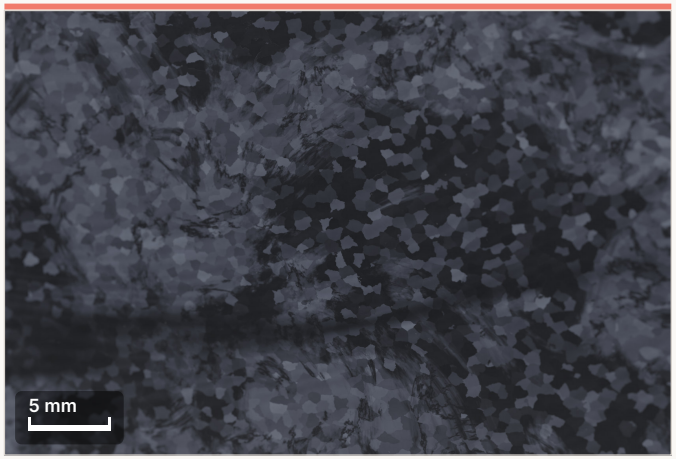
30 cm field	layering and mineral clusters
10 cm field	individual crystals become visible
4 cm field	crystal shapes and microdetail; mean crystal ≈ 0.6 mm



30 CM FIELD 31.2 cm across at the centre



10 CM FIELD 10.3 cm across at the centre



4 CM FIELD (MACRO) 4.1 cm across at the centre

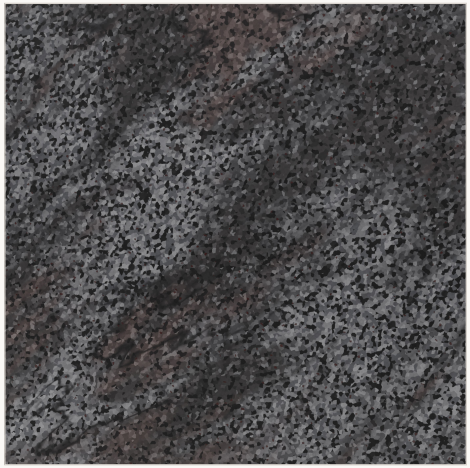
SCALE CHECK

Render vs reference at the same physical size

left: reference · right: v4 render, cropped to the same width

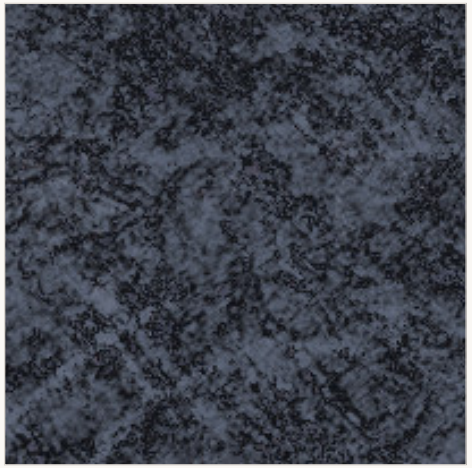
Paradiso 01 · 20.8 cm

PHOTO calibration area

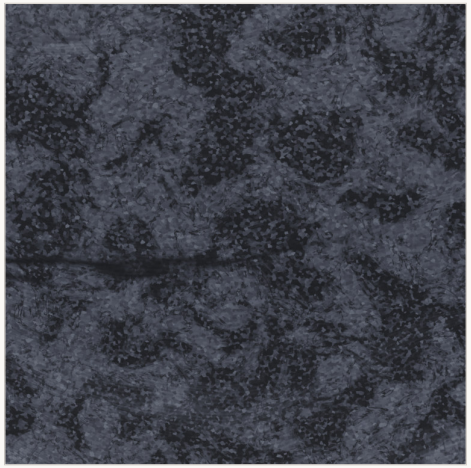


V4 RENDER polished, lit

Bahama Blue · 10 cm

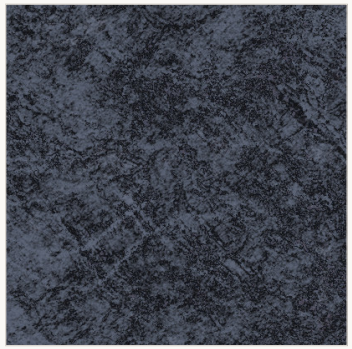


SWATCH 0.49 mm per pixel

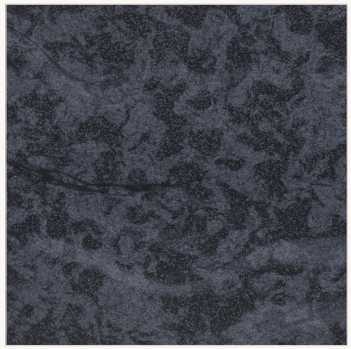


V4 RENDER polished, lit

Bahama Blue · 20 cm



SWATCH



V4 RENDER

Findings

- Paradiso: speckle density, the spacing of the dark biotite clusters and the width of the pale and rust bands match the photo at the same width. The photo shows more elongated, streaky biotite; the render is a little more even.
- Bahama Blue: overall tone and wave spacing match. At 10 cm the render's dark patches are larger and blockier than the swatch's finer lace. That comes from the calibrated pattern (unchanged since v3), not from the new microdetail.
- The renders are lit and glossy and the references are not, so the renders look slightly darker. The unlit albedo matches the references in mean and contrast (earlier calibration reports).

Speckle size, measured

20 CM CROP	REFERENCE	V4 RENDER
Paradiso 01	0.55 mm	0.69 mm
Bahama Blue	0.51 mm	0.72 mm

Half-width of the autocorrelation after removing the large-scale pattern, both images sampled at the reference resolution (0.31 and 0.49 mm per pixel). The render measures 25–40 % coarser. These numbers are close to the pixel size, so treat them as an indication: if the close-ups look too coarse, reduce the preset grain size by 20–30 %. Detail Scale changes only the microdetail, not the crystals.

BEFORE / AFTER

v3 vs v4 at the same scale

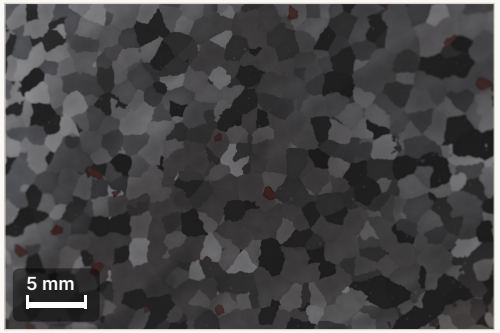
identical camera, light and seed; only the material version differs

Paradiso 01 Salt & Pepper

V3 · 10 CM previous

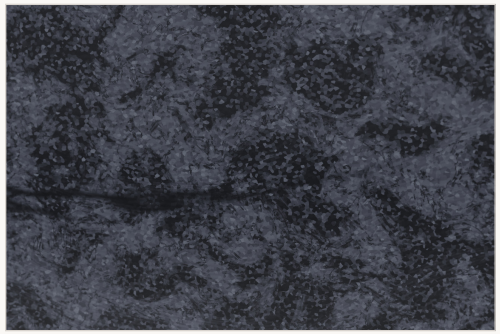
V4 · 10 CM current

V3 · 4 CM previous

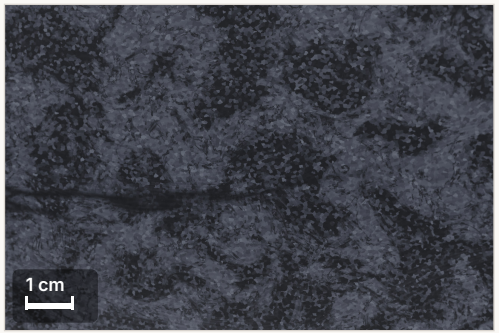


V4 · 4 CM current

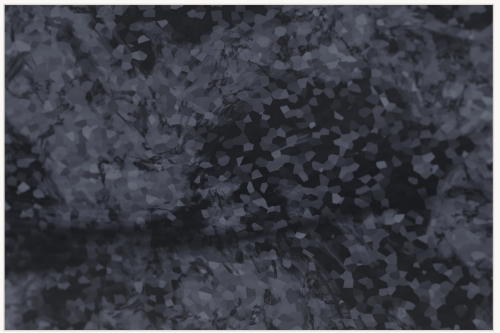
Bahama Blue



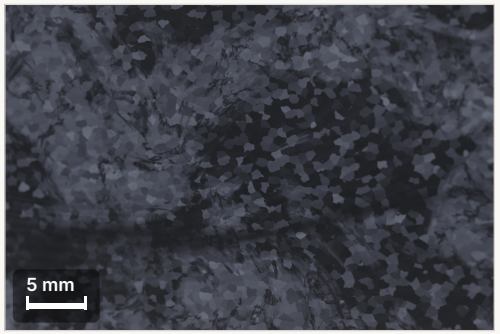
V3 · 10 CM previous



V4 · 10 CM current



V3 · 4 CM previous



V4 · 4 CM current

At 30 cm and beyond the versions are nearly identical

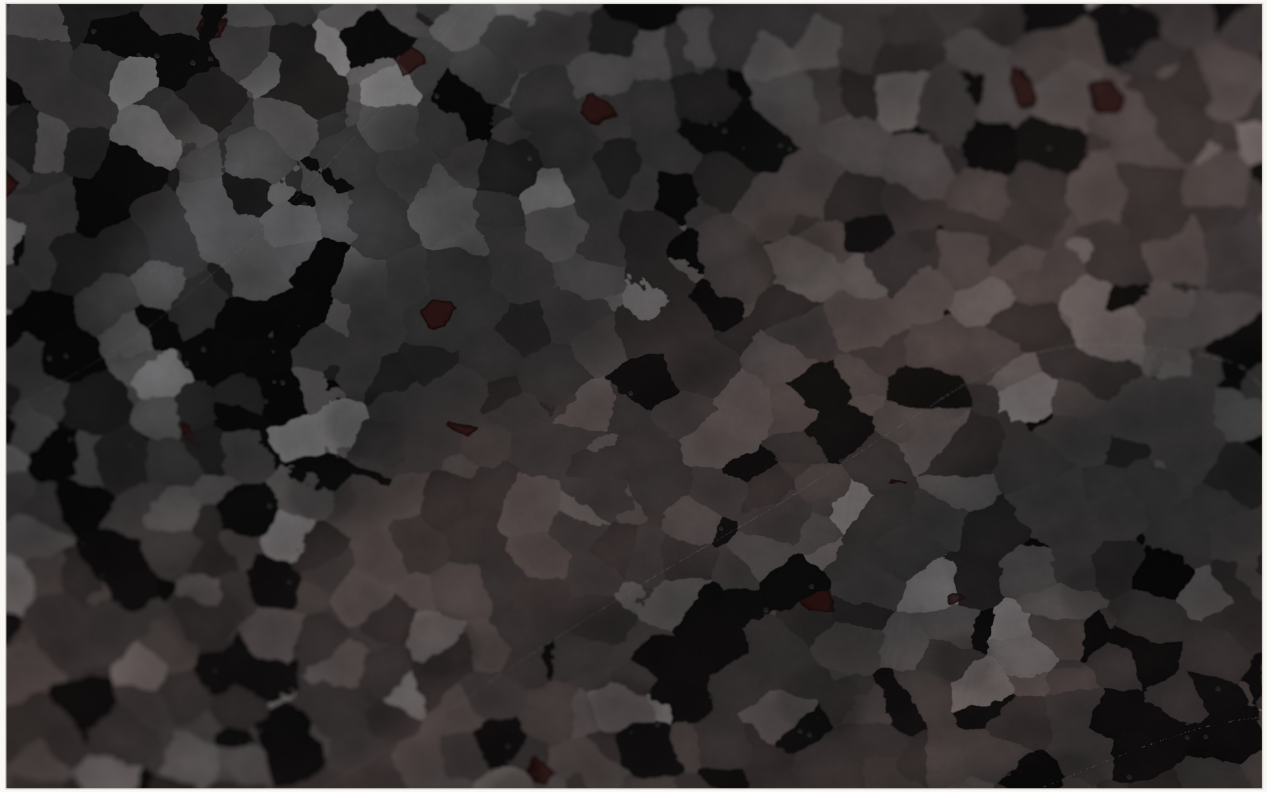
LIT 30 CM RENDER	V3 MEAN / Σ	V4 MEAN / Σ	Δ MEAN
Paradiso 01	0.284 / 0.114	0.289 / 0.106	+1.7 %
Bahama Blue	0.247 / 0.069	0.253 / 0.067	+2.5 %

The changes live below about 2 mm: lobate crystal outlines, internal zoning, garnets, mineral-specific gloss and micrometre relief. At headstone viewing distance they change the overall brightness by only a few percent, mostly from the softer, more varied reflections. Up close the v3 crystals are flat single-colour polygons with straight edges; v4 crystals have irregular outlines, two sizes and internal shading.

PARADISO

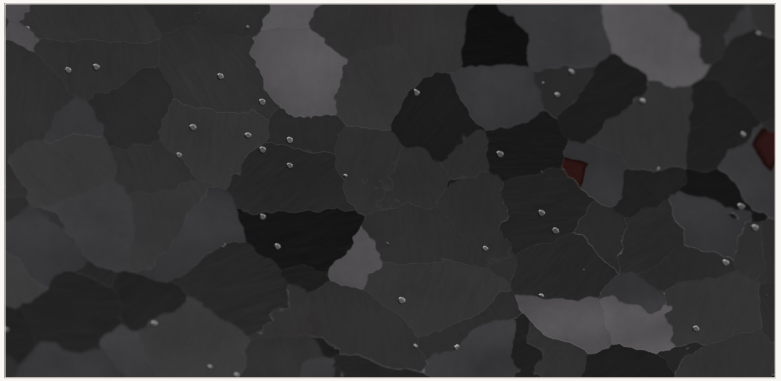
Polished face up close

HD renders · 3200 × 2000 · 512 samples



01 SALT & PEPPER, SOFT LIGHT 40 mm field of view

1:1 CROP about 10 × 8 mm



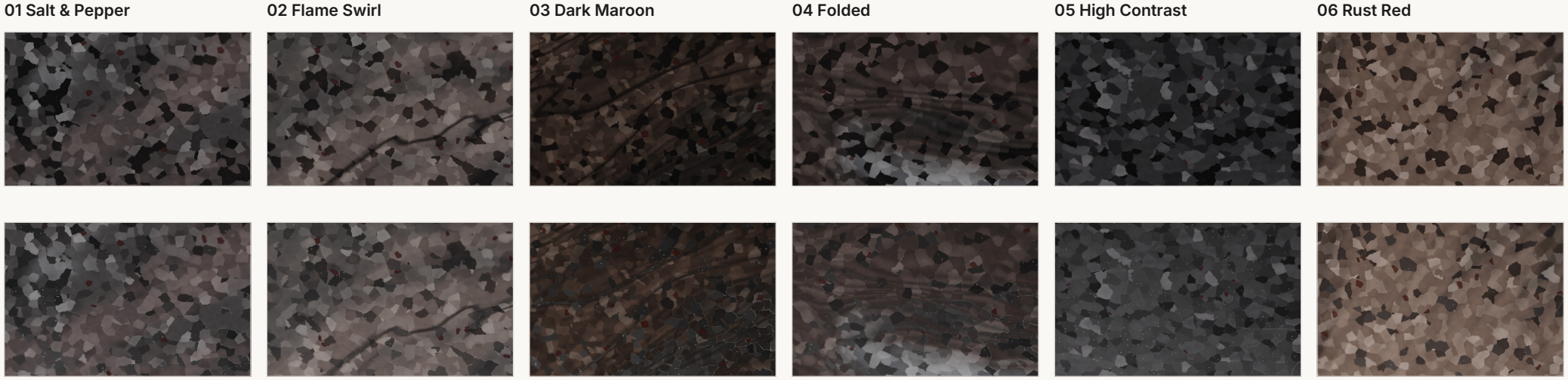
05 HIGH CONTRAST, GRAZING LIGHT crop

Under soft light the crystals now have outline variety, internal shading and the occasional garnet, without changing the overall colour. Under grazing light (light ~10° above the face, opposite the camera) the polished surface stays dark except where rough minerals, pits, boundaries and scratches scatter light, which is the expected response of real polished granite. Relief is in micrometres, so there are no visible bumps or domes.

PARADISO

All six variants

top: soft light · bottom: grazing light · 40 mm field



Per-variant tuning

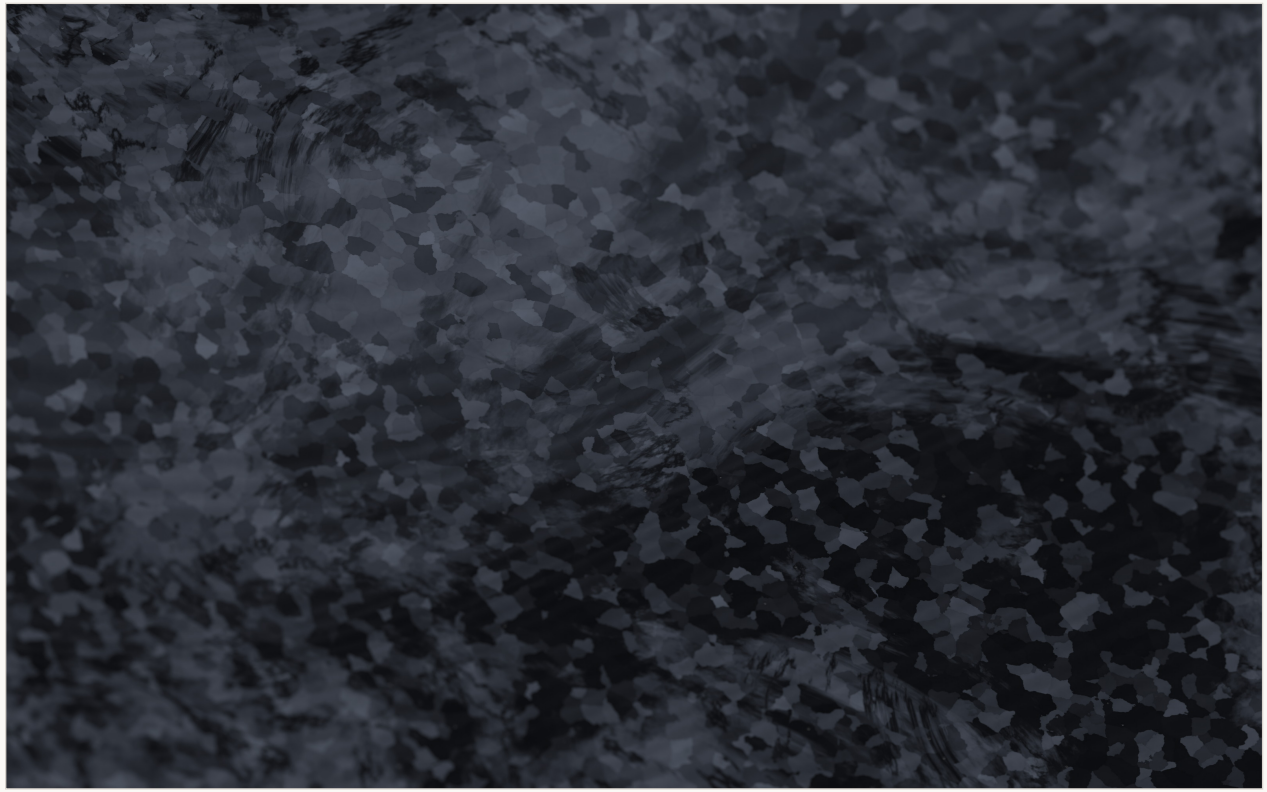
VARIANT	BIOTITE PLUCK	GARNETS	BOUNDARIES	GARNET COLOUR
01 Salt & Pepper	0.50	0.30	0.6	wine red
02 Flame Swirl	0.35	0.45	0.6	wine red
03 Dark Maroon	0.30	0.55	0.5	darker, browner
04 Folded	0.45	0.35	0.7	wine red
05 High Contrast	0.60	0.35	0.6	wine red
06 Rust Red	0.30	0.60	0.5	warmer red

Biotite-rich variants (01, 05) get more plucking; the garnet-rich warm variants (03, 06) get more garnets. Crystal shapes are shared (irregularity 0.5, size variation 0.35); everything else follows each variant’s calibrated preset.

BAHAMA BLUE

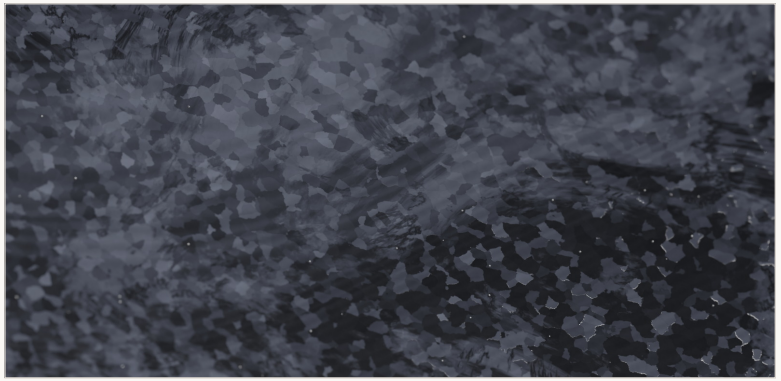
Polished face up close

fine-grained stone: a different micro graph



BAHAMA BLUE, SOFT LIGHT HD, 40 mm field of view

BEFORE (V3) soft light



AFTER, GRAZING LIGHT

Why it differs from Paradiso

- Crystals are under 1 mm, so large facet flicker would look wrong; a 0.15 mm roughness sparkle replaces it.
- The dark network is fine, well-bonded biotite/hornblende: only +0.10 roughness and a 0.8 µm recess.
- No garnets or twin lamellae are visible in the references, so none are added.

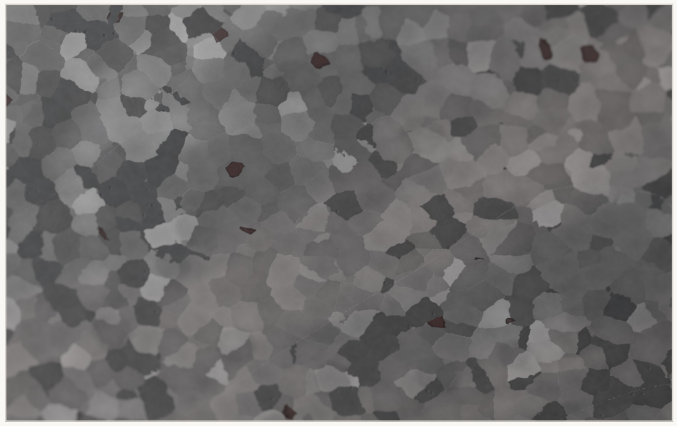
Result

- The polished face reads as a dense, uniform gloss with soft crystal fuzz and a faint darker grain mosaic.
- Waves and the lace-like network from the calibrated pattern are unchanged.
- Under grazing light only sparse rough grains and rare pits catch the light.

EDGE FINISHES

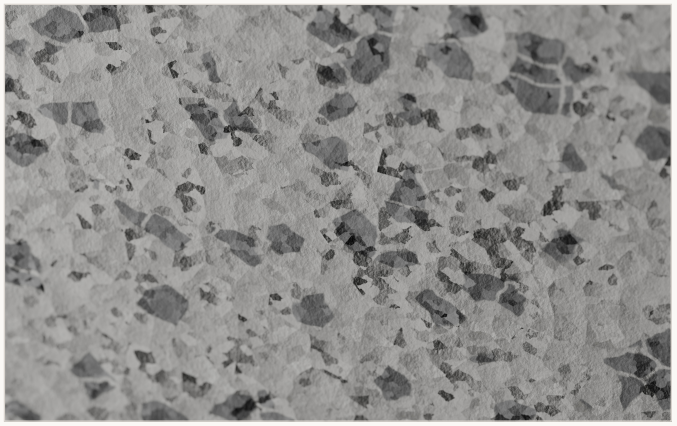
Honed, rock-pitched and flamed

raking light ~8° above the surface, 90° to the view

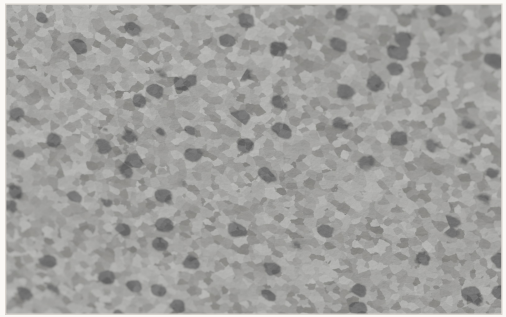


PARADISO HONED matte, 4× relief, crystal-level polish

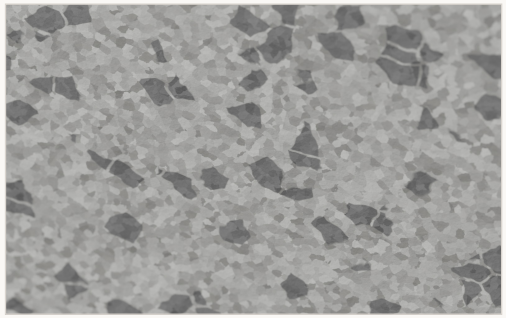
PARADISO 04 ROCK-PITCHED soft light, 0.25 mm spall steps



BAHAMA BLUE FLAMED HD, spalls, craters, mica glints



FLAMED SPECKS BEFORE round dots



AFTER angular grains

Flamed calibration (10 cm patch)

	MEAN SRGB	Σ	P5–P95
Photo rim	187 186 181	31	131–231
v3 shader	188 187 183	28	127–224
v4 shader	184 183 179	31.2	118–224

Specks now cover the same area as before but follow mineral grains, so the flamed edge no longer looks polka-dotted at macro scale, and its contrast is now closer to the photo.

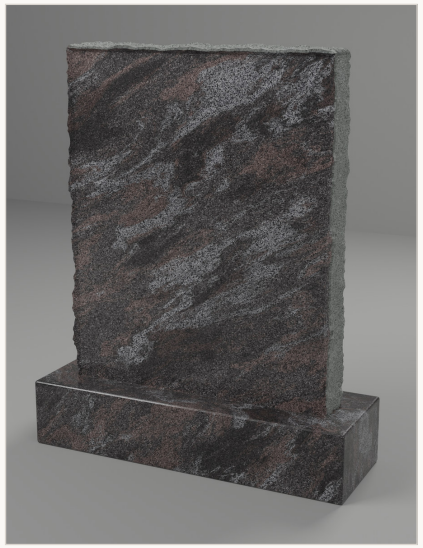
All seven materials on a headstone (v4)

studio stage · 62 × 78 cm slab · polished face, honed sides

01 SALT & PEPPER

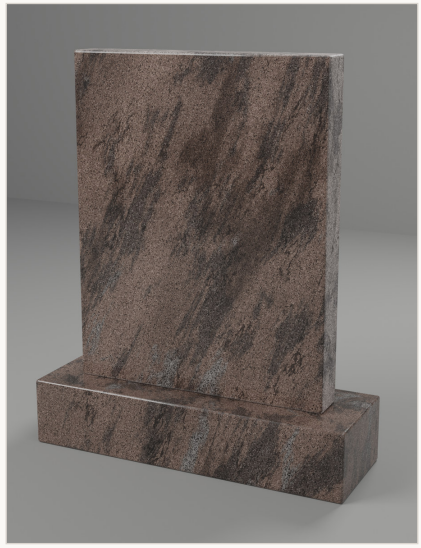
02 FLAME SWIRL

03 DARK MAROON

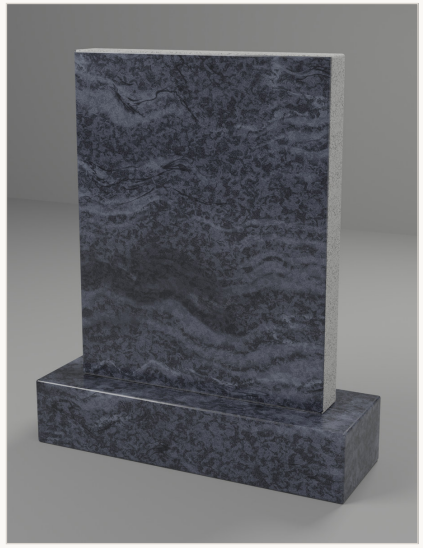


04 FOLDED ROCK

05 HIGH CONTRAST



06 RUST RED



07 BAHAMA BLUE

BEFORE / AFTER

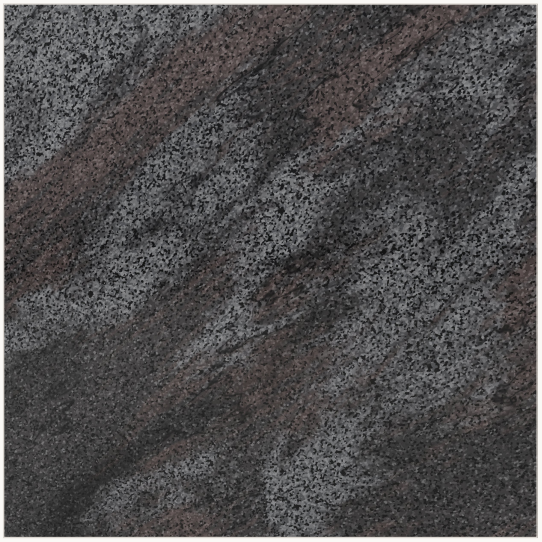
Headstone renders: previous (v3) vs current (v4)

same stage, camera, lights and seed

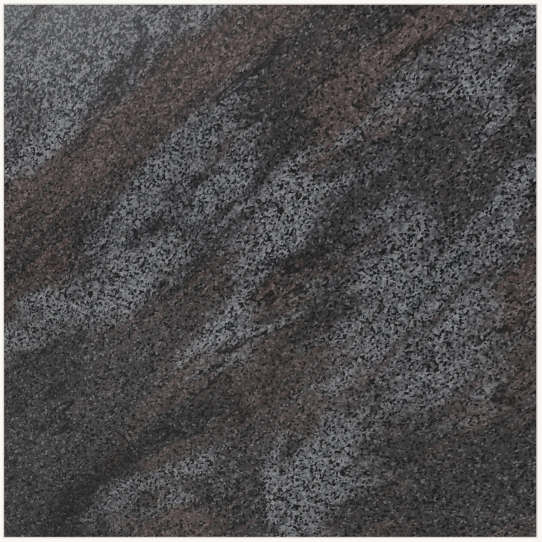
01 Salt & Pepper

V3 · HEADSTONE

V4 · HEADSTONE

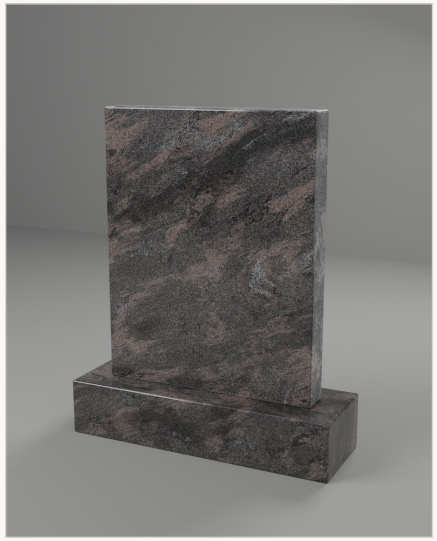


V3 · CLOSE-UP

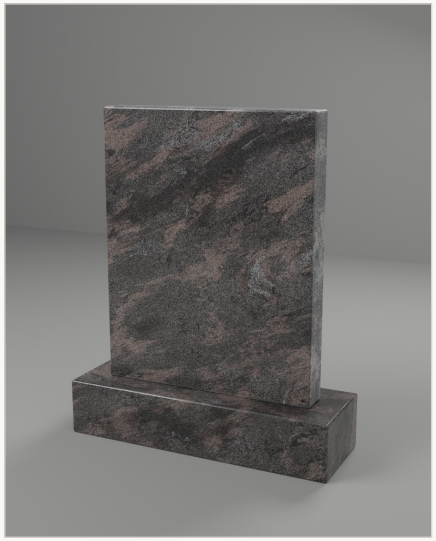


V4 · CLOSE-UP

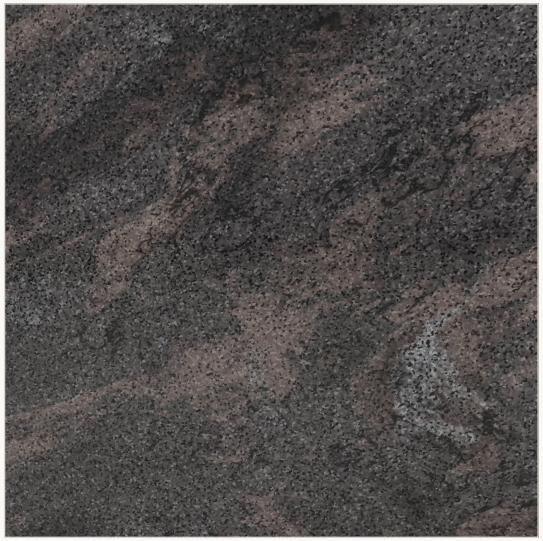
02 Flame Swirl



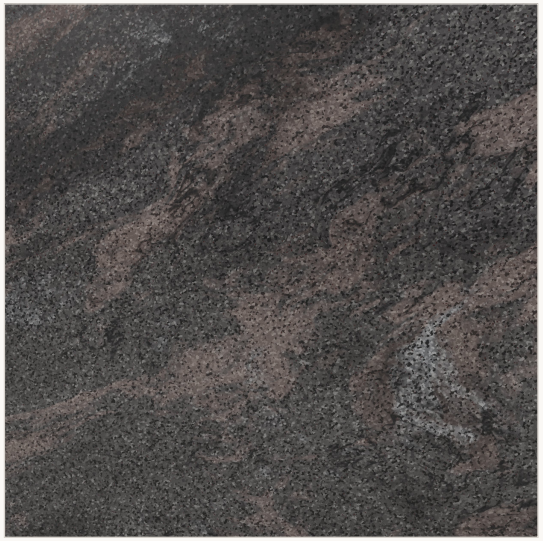
V3 · HEADSTONE



V4 · HEADSTONE



V3 · CLOSE-UP



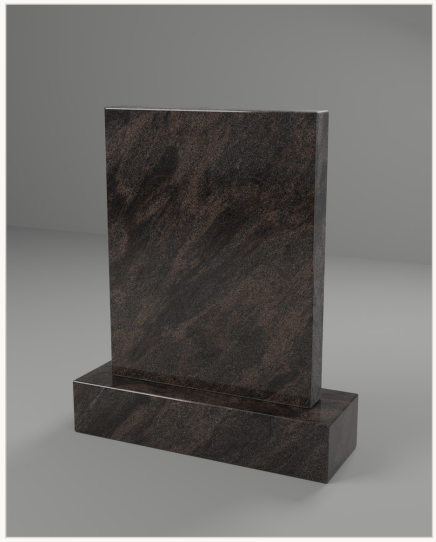
V4 · CLOSE-UP

BEFORE / AFTER

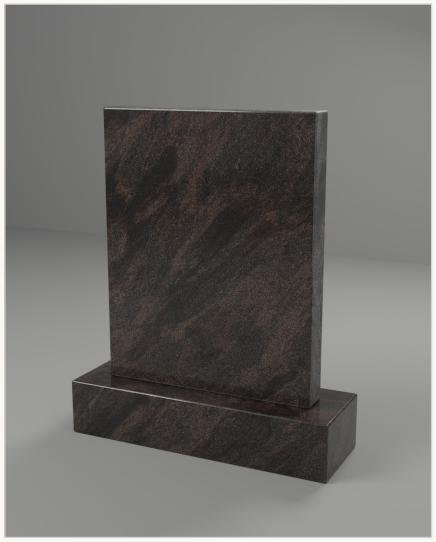
Headstone renders: previous (v3) vs current (v4) (cont.)

same stage, camera, lights and seed

03 Dark Maroon



V3 · HEADSTONE

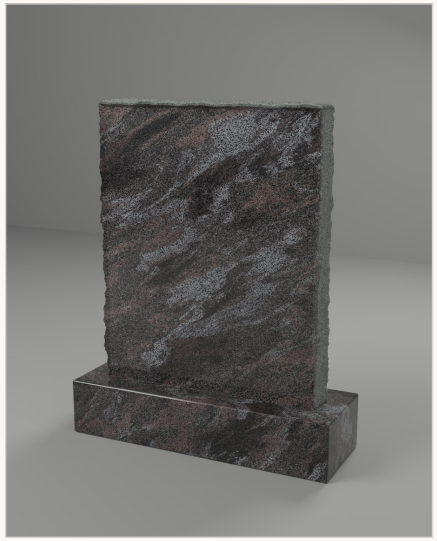


V4 · HEADSTONE

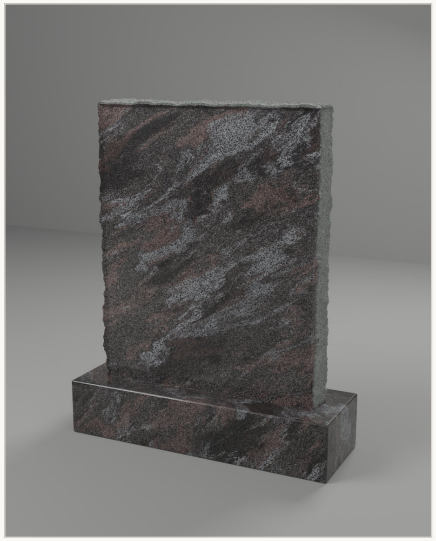
V3 · CLOSE-UP

V4 · CLOSE-UP

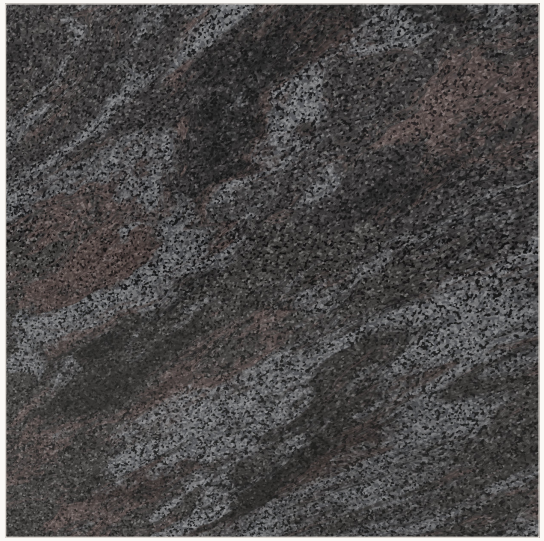
04 Folded Rock



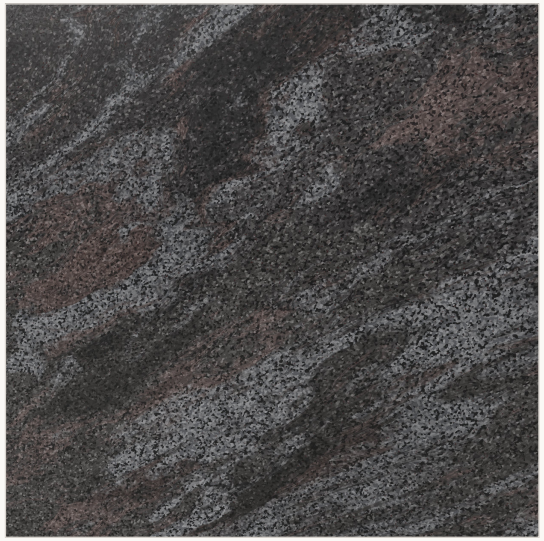
V3 · HEADSTONE



V4 · HEADSTONE



V3 · CLOSE-UP



V4 · CLOSE-UP

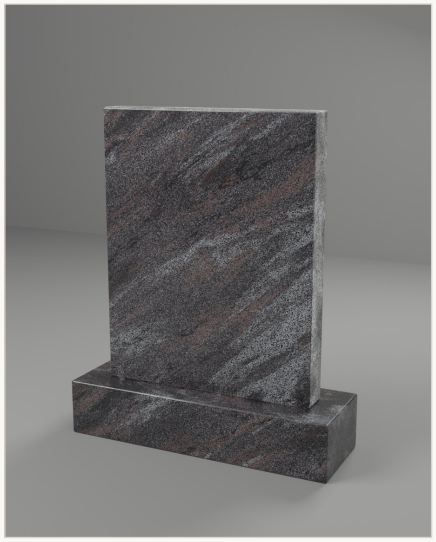
BEFORE / AFTER

Headstone renders: previous (v3) vs current (v4) (cont.)

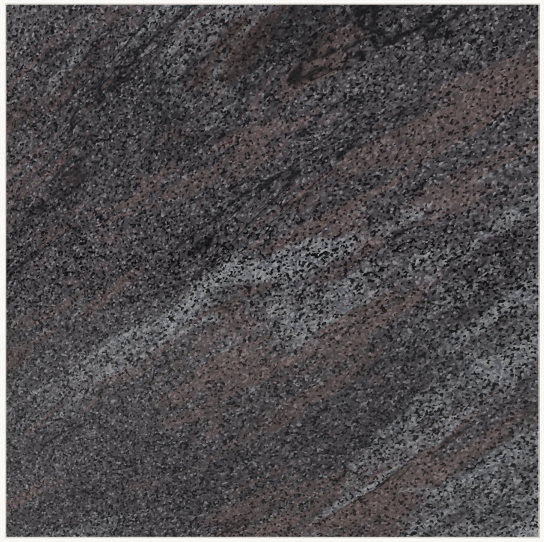
same stage, camera, lights and seed

05 High Contrast

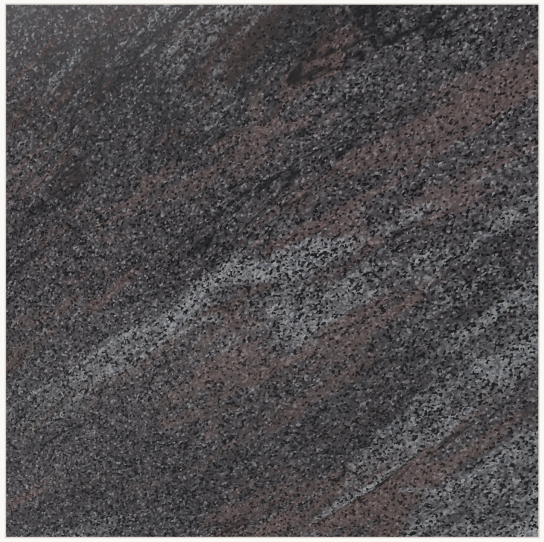
V3 · HEADSTONE



V4 · HEADSTONE

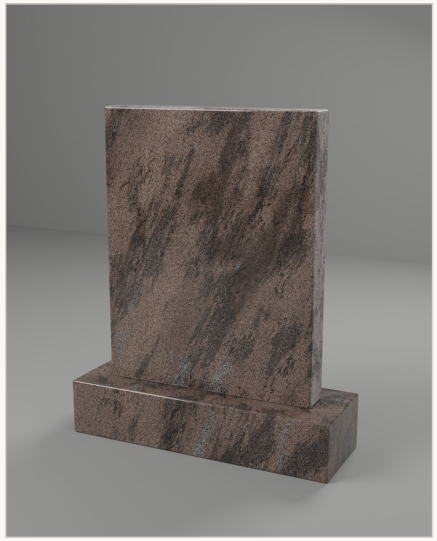


V3 · CLOSE-UP

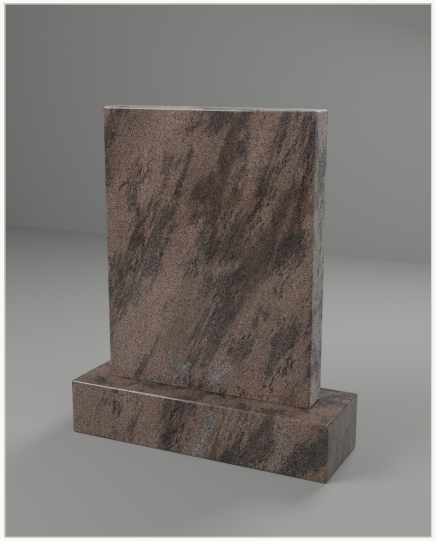


V4 · CLOSE-UP

06 Rust Red

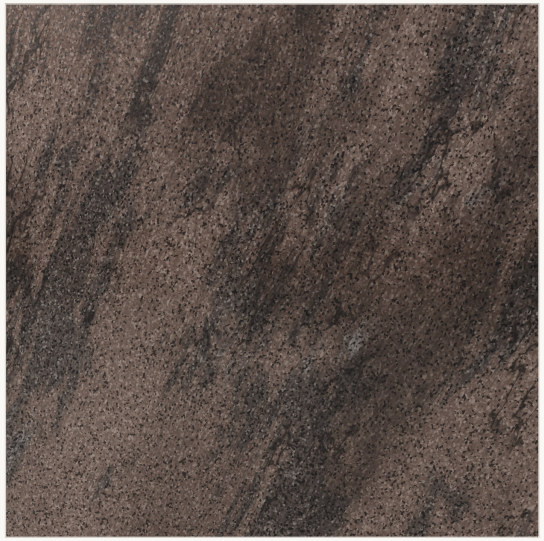


V3 · HEADSTONE



V4 · HEADSTONE

V3 · CLOSE-UP

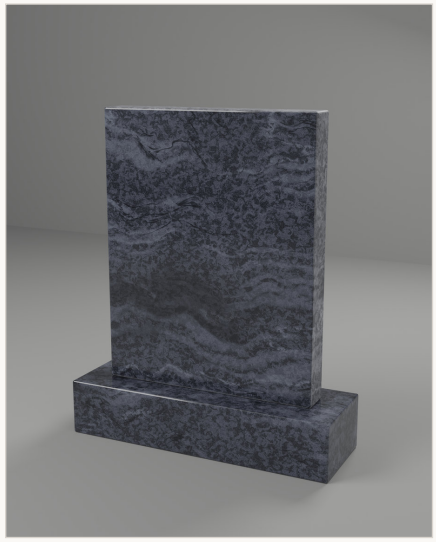


V4 · CLOSE-UP

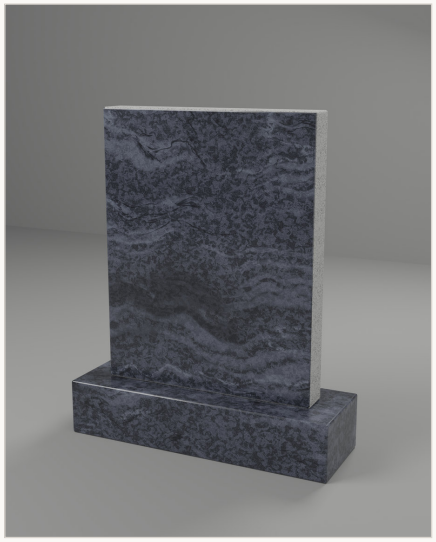
Headstone renders: previous (v3) vs current (v4) (cont.)

same stage, camera, lights and seed

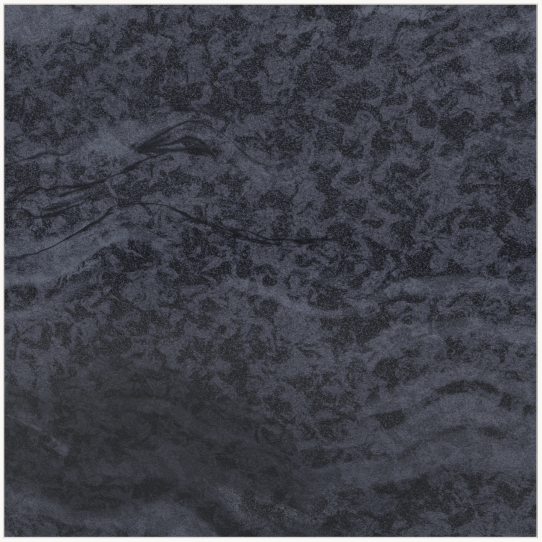
07 Bahama Blue



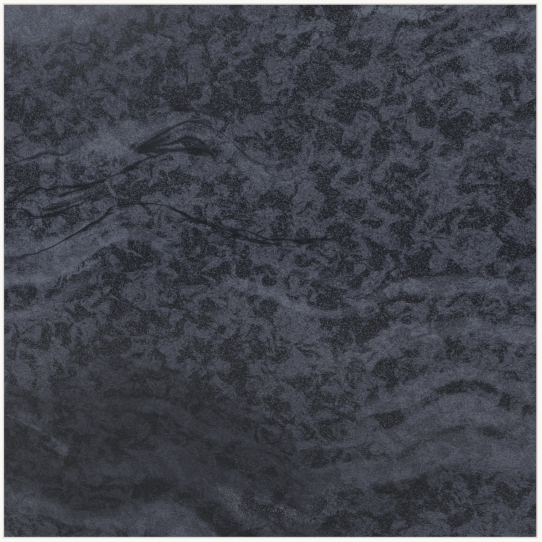
V3 · HEADSTONE



V4 · HEADSTONE



V3 · CLOSE-UP



V4 · CLOSE-UP

What changes on a headstone

Measured luminance (lit renders)

MATERIAL	HEADSTONE V3	V4	Δ	CLOSE-UP V3	V4	Δ
01 Salt & Pepper	0.288	0.291	+0.9 %	0.286	0.290	+1.4 %
02 Flame Swirl	0.286	0.289	+1.0 %	0.286	0.290	+1.2 %
03 Dark Maroon	0.190	0.194	+2.2 %	0.188	0.195	+3.4 %
04 Folded Rock	0.280	0.283	+1.1 %	0.274	0.279	+1.6 %
05 High Contrast	0.291	0.295	+1.3 %	0.290	0.295	+1.9 %
06 Rust Red	0.317	0.317	+0.0 %	0.305	0.306	+0.3 %
07 Bahama Blue	0.252	0.256	+1.8 %	0.250	0.256	+2.2 %

Headstone luminance is measured over the centre of the polished face in the 2400 × 3000 beauty render; close-up over the whole 2400 × 2400 detail render. Both versions were rendered with the same scene file (hd_render.py).

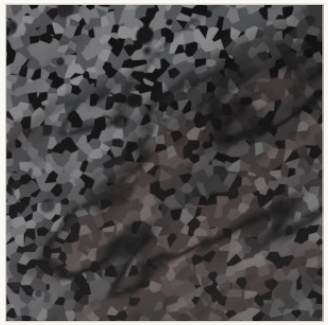
Same at viewing distance

- Pattern, colour, banding and the position of every feature are unchanged: v4 reuses the calibrated pattern graph and the same seeds.
- The face is 0–3 % brighter in v4. This comes from the new crystal shading and mineral-specific gloss, and is smaller than a typical change in lighting.

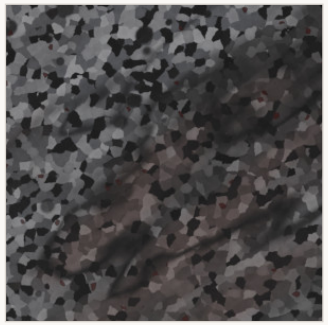
Different up close

- In the close-ups (about 40 cm across) v4 shows crystal-to-crystal gloss changes and softer crystal edges.
- At the 4 cm macro view the difference is large: see the scale and macro pages.
- Rock-pitched and flamed finishes change the most, because their relief was rebuilt (see Edge finishes).

Close-up renders at 1:1 (each square about 5 cm of stone)

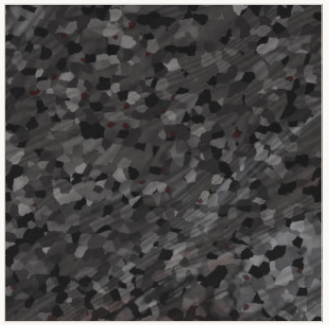


V3 · SALT & PEPPER

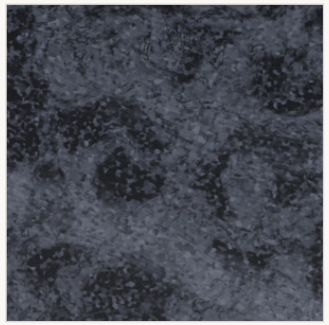


V4 · SALT & PEPPER

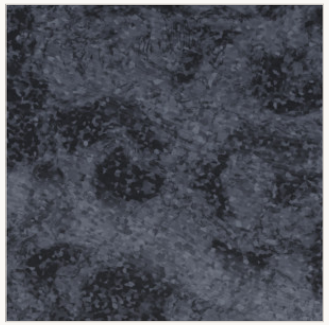
V3 · FOLDED ROCK



V4 · FOLDED ROCK



V3 · BAHAMA BLUE

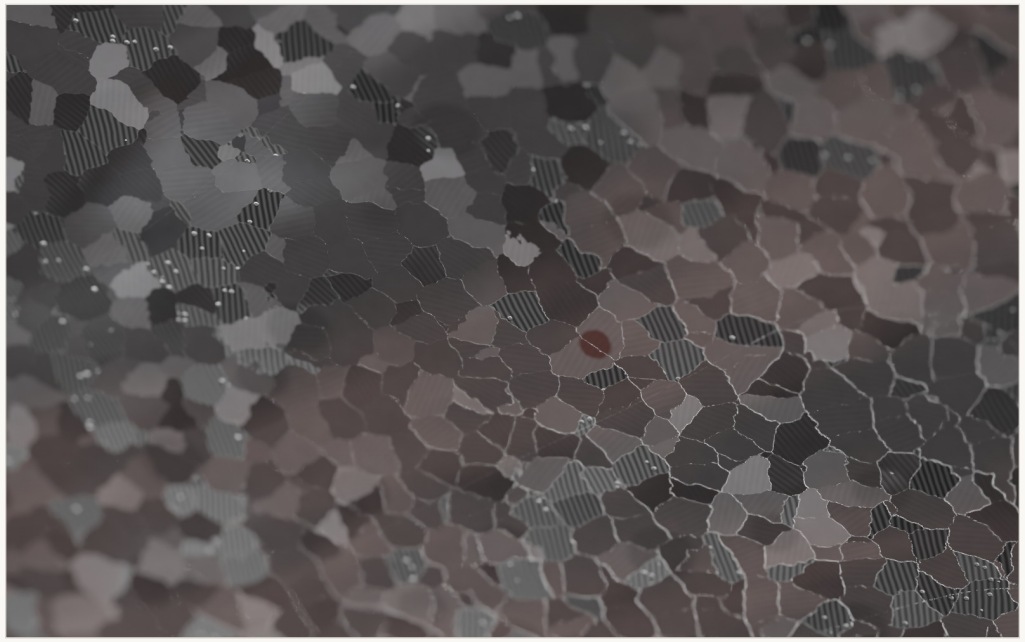


V4 · BAHAMA BLUE

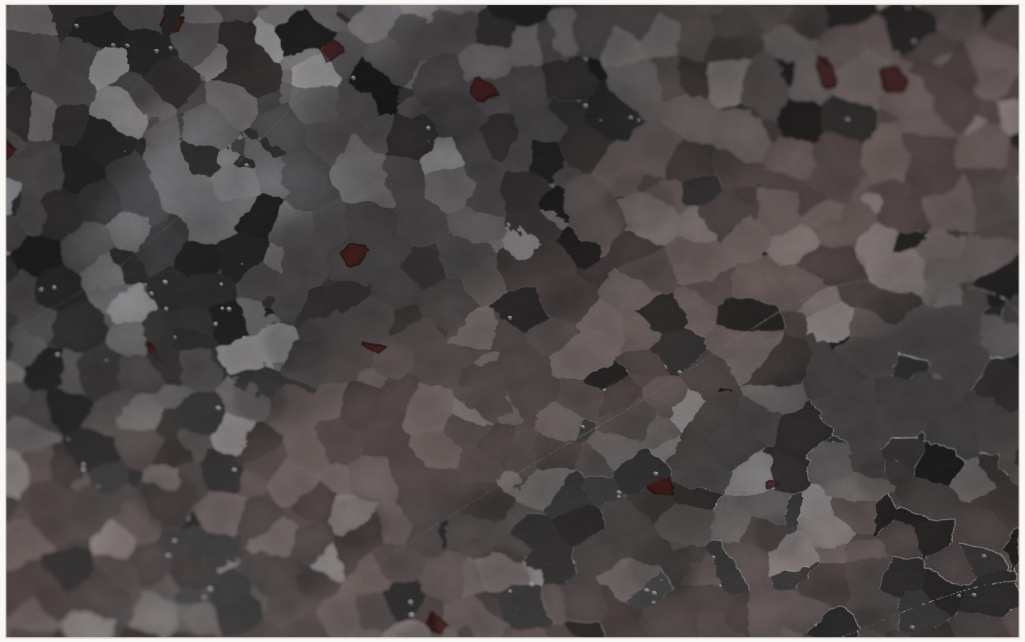
VALIDATION

What the grazing-light tests caught

first pass (left) vs final (right), grazing light



FIRST PASS bright "cracked" web, striped mica, round garnet



FINAL hairlines only, irregular sheen, angular garnets

ARTEFACT	FIX
Grain boundaries glowed as a bright web	recess 2 → 0.7 µm, roughness +0.12 → +0.05
Bright rims around biotite (step edges)	biotite recess 4 → 1.5 µm
Regular "corduroy" stripes on mica	periodic wave → stretched distorted noise, half strength
Too many plucked pits (pitted look)	pit density ×0.55, depth 18 → 10 µm
Twin lamellae read as wallpaper	60 % → 35 % of crystals, faded in patches, half amplitude
Garnet was a soft round dot	whole angular grains with a dark rim
Flamed specks were polka dots	angular grain-shaped specks, same coverage

USING IT

Controls and macro render settings

Blender/Granite_Macro_v4.blend

Microdetail node controls

CONTROL	DEFAULT	WHAT IT DOES
Detail Scale	1	finer pits, scratches, spalls (>1)
Bump Strength	1	1 = physically scaled relief
Roughness Variation	1	0 = uniform polish
Facet Tilt	1	crystal-to-crystal flicker
Crystal Texture	1	zoning, mottle, lamellae
Grain Boundaries	0.4–0.7	hairline cracks
Polish Scratches	0.2–0.25	scratch sets (polished only)
Polish Waviness	1	slab waviness in reflections
Biotite Pluck	0.1–0.6	plucked pits in dark minerals
Garnet Inclusions	0.3–0.6	Paradiso only
Fine Sparkle / Mica Glints	1 / 0.5	Bahama Blue only
Flamed Relief	1	Bahama Blue flamed spall depth

Seed: one SEED value node per material drives both groups.

Final macro render settings (scene MACRO_STAGE)

- Cycles, GPU (OptiX), 1024 samples, adaptive threshold 0.005, OpenImageDenoise.
- Glossy bounces 6, indirect clamp 10, pixel filter 1.2, AgX view.
- Camera within ~30° of the face normal; f/22–32 at 15–20 cm focus.
- Give polished faces something to reflect: a large dim card or an HDRI.
- Polished and honed faces need no subdivision (µm relief via bump). Rock-pitched faces need subdivision (adaptive, or ~6 levels) for their displacement.

Recovering the originals

- *_ORIG materials and GRANITE_*_ORIG groups are stored in the .blend with a fake user.
- The v3 scripts are in Blender/backup_v3_scripts; the folder scripts now build v4.
- Test renders: Blender/macro_tests; stage script: Blender/macro_stage.py.